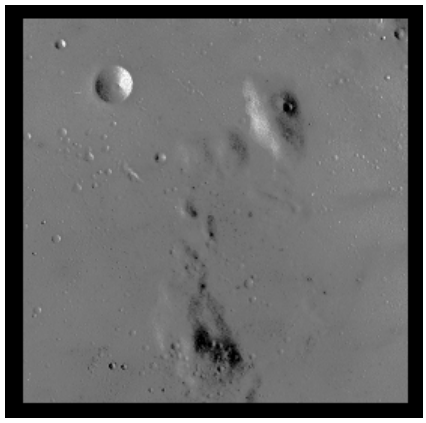
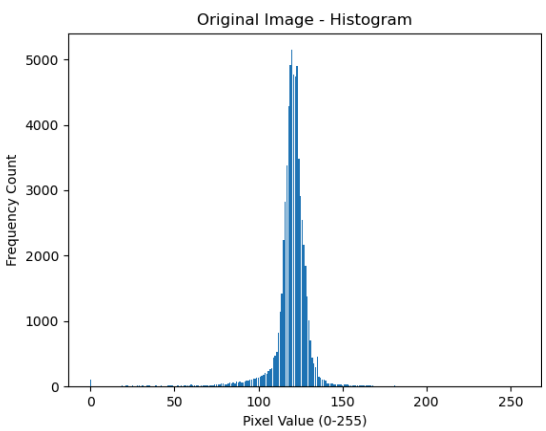
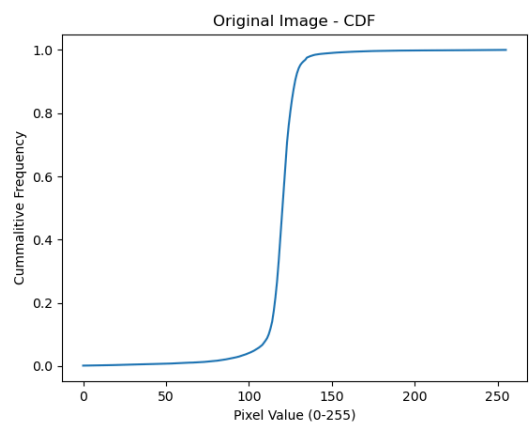
Nolan Knight
MSAI495 - Computer Vision
MP#2
April 12, 2026

In this problem, we applied histogram equalization to a sample image to better distribute the intensities found within the image. The following sample image was used for this problem:



As shown in the image, it is hard to distinguish many of the image properties, as many of the intensities of the grayscale image are similar. In order to apply histogram equalization, we had to calculate the histogram of the pixel intensities within the image, or count the frequency that each intensity occurred in the image. After this, we could derive the cumulative distribution for each intensity (CDF), by calculating the running average and dividing by the total number of pixels in the image. This gives us a probability curve of the likelihood of each pixel intensity in the image. This produced the following histogram and cumulative distribution:





Using this histogram and CDF, we can get the transformation of each pixel in the output image by applying the Histogram Equalization algorithm equation from slide 6:

$$T(r_k) = \sum_{j=0}^{k} P(r_j) \ * \ L2$$

This equation maps each pixel intensity in the original image to a new intensity value by using the cumulative probability up to that intensity level. The result is then scaled by the L2, where L2 is the total number of possible intensity levels. This process normally redistributes pixel intensities to enhance the overall contrast of the image. After applying this function to the

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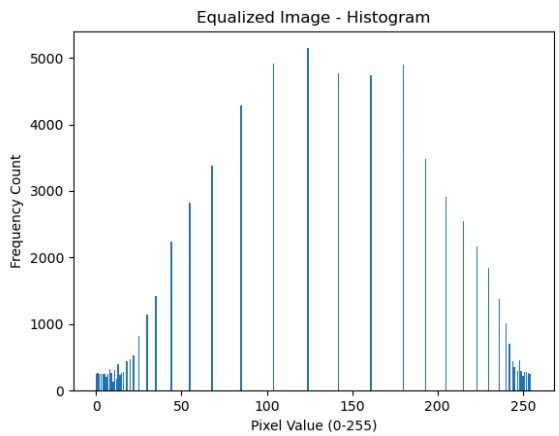
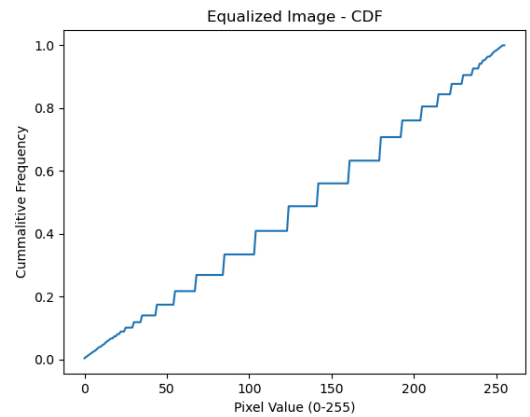
input image for each pixel intensity, the output image was transformed into the following intensity histogram and cumulative distribution:





As the updated histogram and CDF of the output image represents, the intensities of the output image are nearly uniformly distributed. This is shown, as the histogram appears to have a more equal distribution of intensities, spanning the whole range from 0-255, and the CDF is nearly linear suggesting that there is uniform distribution. As a result, the output image after the applied histogram equalization was:

As shown in the image, now that there is a much better distribution of gray scale intensities, it is much easier to tell what is going on in the image.

EC: Light Filtering

Using the matrix light filtering tactics discussed in class, I also applied linear and quadratic light filtering to the adjusted image for the extra credit problems. This was done by calculating

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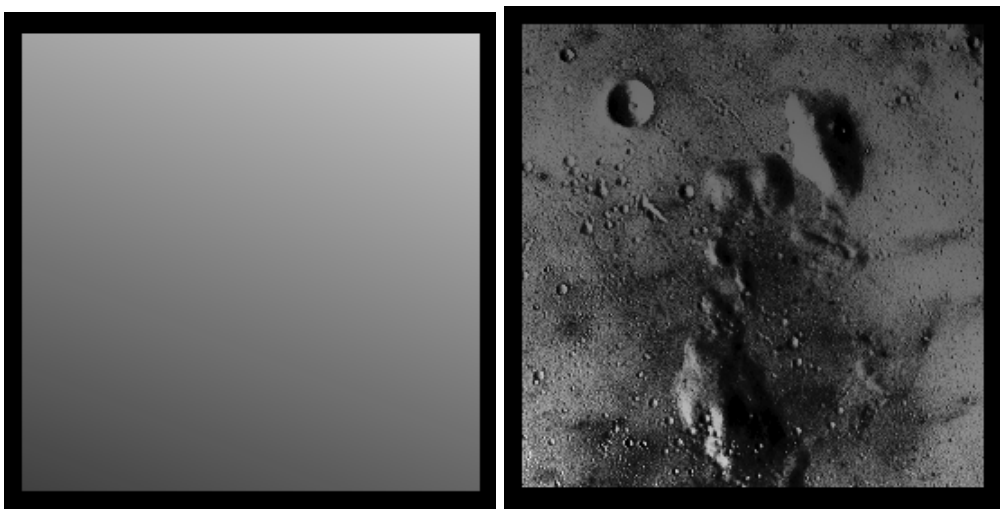
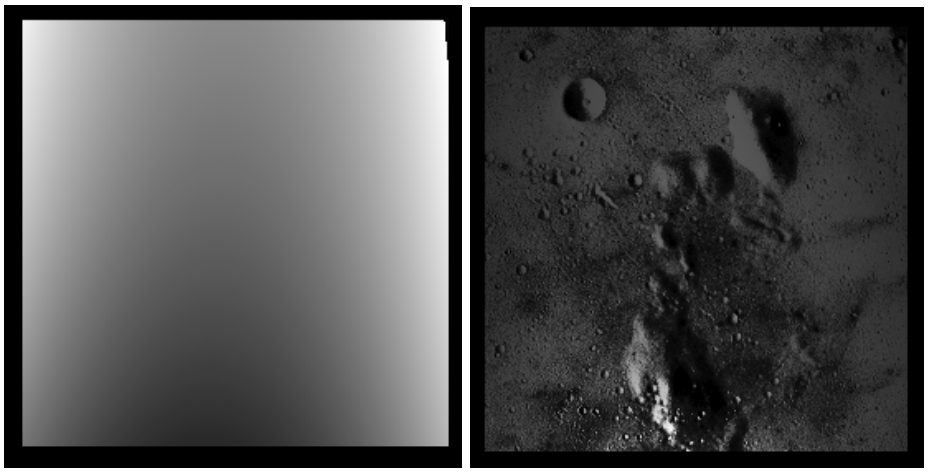
the A matrix using spatial information, and the y matrix which is each pixel. Using the pseudo inverse of A, we can derive the state vector of the linear and quadratic system to form a light adjustment filter. Using this method, the following output was obtained for the linear and quadratic filters:

Linear:



As seen from the output, the image is brighter in the top corner, and this is reflected in the light map generated. This resulted in an output with a much better light distribution in the image.

Quadratic:



As seen from the output, the image is brighter on the outskirts and darker in the center, which is captured by the quadratic light filter. This resulted in an output with a much better light distribution in the image across the whole image, by darkening the sides and lightening the center.